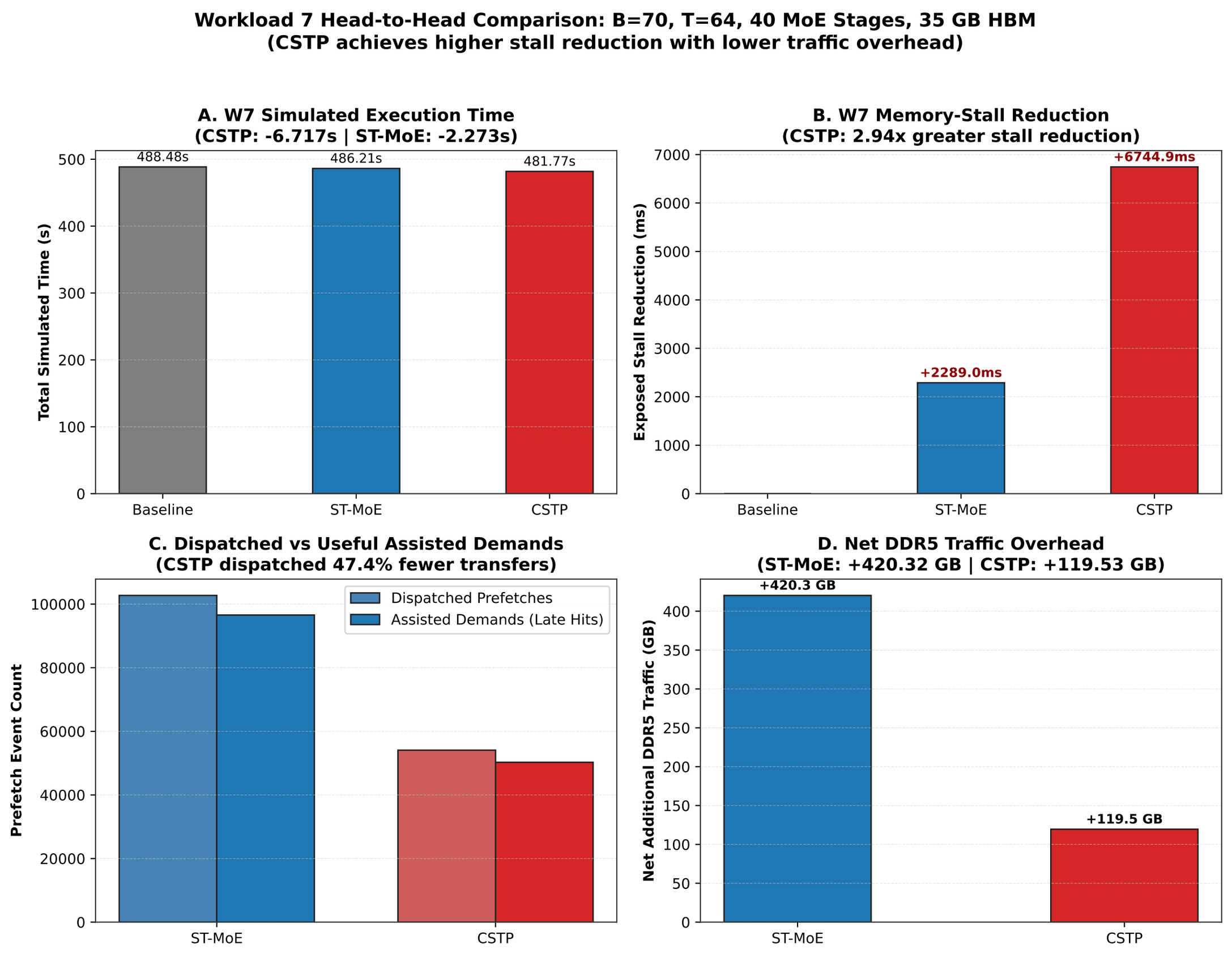
Workload 7 Head-to-Head Comparison: B=70, T=64, 40 MoE Stages, 35 GB HBM
(CSTP achieves higher stall reduction with lower traffic overhead)
A. W7 Simulated Execution Time
(CSTP: -6.717s | ST-MoE: -2.273s)
Total Simulated Time (s)
488.48s
486.21s
481.77s
Baseline
ST-MoE
CSTP
B. W7 Memory-Stall Reduction
(CSTP: 2.94x greater stall reduction)
Exposed Stall Reduction (ms)
+2289.0ms
+6744.9ms
C. Dispatched vs Useful Assisted Demands
(CSTP dispatched 47.4% fewer transfers)
Prefetch Event Count
Dispatched Prefetches
Assisted Demands (Late Hits)
D. Net DDR5 Traffic Overhead
(ST-MoE: +420.32 GB | CSTP: +119.53 GB)
Net Additional DDR5 Traffic (GB)
+420.3 GB
+119.5 GB